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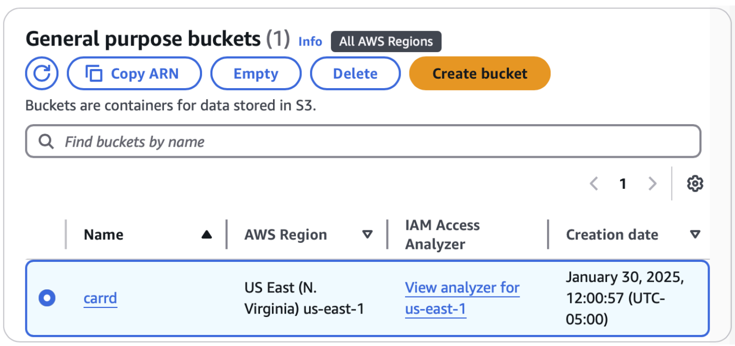
MCY 611
Lab 2 AWS S3

MCY 611 Cloud Computing
Lab 2: AWS S3

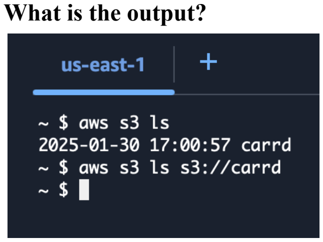
Login to the AWS management console as the root user with your NKU email and AWS account password. After logging in, launch the AWS CloudShell terminal by clicking on the terminal icon next to the magnifying glass.

Part 1: Basic Operations with S3

1. In the terminal, make a bucket via the following mb command:
aws s3 mb s3://*your_user_name*
What is the output?

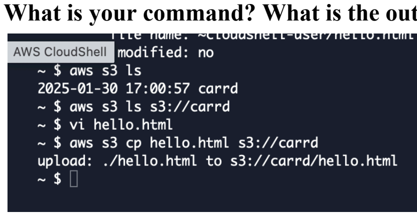


2. List all buckets owned by the user via the ls command.
aws s3 ls
What is the output?



3. List the contents of the bucket via the following ls command:
aws s3 ls s3://*your_user_name*
What is the output?

4. Use vi or other text editor to create a simple html file called hello.html. The content of the file is as follows:
<html><body><h1>Amazon S3</h1>
Hello World!
</body></html>
Upload the file into the bucket and make the file available to the public via the following command:
aws s3 cp hello.html s3://*your_user_name*
What is your command? What is the output?

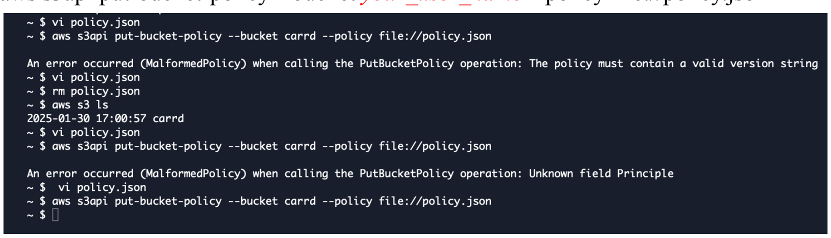


5. Delete the public access block.
aws s3api delete-public-access-block --bucket *your_username*

6. Set bucket policy to allow public read access. First, create a file called policy.json with the following contents:

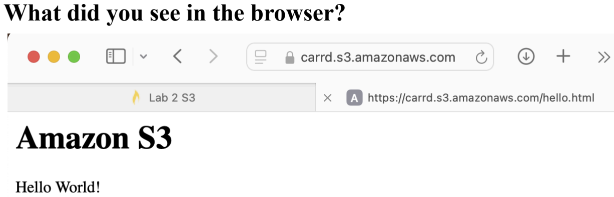
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::your_user_name/*"
            ]
        }
    ]
}
```

Set the bucket policy with this command:
aws s3api put-bucket-policy --bucket *your_user_name* --policy file://policy.json



7. Amazon S3 is object storage over the Internet. The S3 object is accessible over the Internet. Open a browser and access to https://*your_user_name*.s3.amazonaws.com/hello.html.

What did you see in the browser?



Part 2: Static Website Hosting with S3

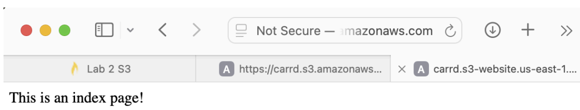
8. Let's enable the bucket for static website hosting. Create a directory called website with two html files in the directory. One file is named index.html. Its content is as follows:
<html><body>
This is an index page!
</body></html>
The other file is named error.html. Its content is as follow:
<html><body>
Sorry, we can't find that page!
</body></html>

```
~ $ aws s3api put-bucket-policy
~ $ mkdir website
~ $ cd website
website $ vi index.html
website $ vi error.html
```

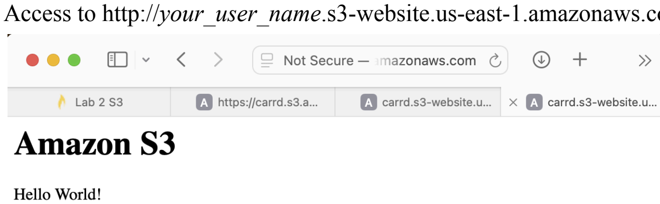
The sync command compares the source directory with your S3 bucket and uploads only new or changed files. Please upload the files by running the following command from within the website directory:
aws s3 sync ./ s3://*your_user_name*/
What is your command and its output?

```
website $ aws s3 sync ./ s3://carrd/
upload: ./error.html to s3://carrd/error.html
upload: ./index.html to s3://carrd/index.html
website $
```

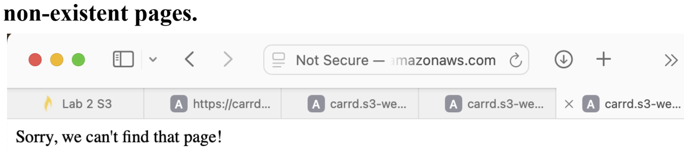
The following is how to enable the bucket for static website hosting:
aws s3 website s3://*your_user_name*/ --index-document index.html --error-document error.html
Open a browser and access to http://*your_user_name*.s3-website.us-east-1.amazonaws.com/ . **What did you see in the browser? Why?**
website $ aws s3 website s3:/*carrd*/ --index-document index.html --error-document error.html is configured as the default page for my S3 static website.



Access to http://*your_user_name*.s3-website.us-east-1.amazonaws.com/hello.html . **What did you see in the browser?**



Access to http://*your_user_name*.s3-website.us-east-1.amazonaws.com/2.html . **What did you see in the browser? Why?**
website $ aws s3 website s3:/*carrd*/ --index-document index.html --error-document error.html displays this for non-existent pages.



9. Create a web page redirection rule and add metadata to an object via the following s3api command:
aws s3api put-object --bucket *your_user_name* --key hello.html --website-redirect-location http://www.nku.edu/ --metadata redirection_creator=mcy611

```
website $ aws s3api put-object --bucket carrd --key hello.html --website-redirect-location http://www.nku.edu/ --metadata redirection_
creator=mcy611
{
    "ETag": "\"d41d8cd98f00b204e9800998ecf8427e\"",
    "ChecksumCRC64NVME": "AAAAAAAAAAA=",
    "ChecksumType": "FULL_OBJECT",
    "ServerSideEncryption": "AES256"
}
website $
```

Access to http://*your_user_name*.s3-website.us-east-1.amazonaws.com/hello.html . **What did you see in the browser? Why?**
The NKU Direct Admit website
The put-object command is configured with the --website-redirect-location parameter which redirects any request made to the hello.html object stored in the s3 bucket to the www.nku.edu website.

10. Retrieve an object's metadata via the following:

aws s3api head-object --bucket *your_user_name* --key hello.html

What is the output?

```
website $ aws s3api head-object --bucket carrd --key hello.html
{
    "AcceptRanges": "bytes",
    "LastModified": "2025-01-30T18:42:09+00:00",
    "ContentLength": 0,
    "ETag": "\"d41d8cd98f00b204e9800998ecf8427e\"",
    "ContentType": "binary/octet-stream",
    "WebsiteRedirectLocation": "http://www.nku.edu/",
    "ServerSideEncryption": "AES256",
    "Metadata": {
        "redirection_creator": "mcy611"
    }
}
website $
```

Part 3: Cleanup

11. Remove an object, the index page, via the following command:

aws s3 rm s3://*your_user_name*/index.html

What is the output?

```
website $ aws s3 rm s3://carrd/index.html
delete: s3://carrd/index.html
website $
```

12. Remove the bucket via the following:

aws s3 rb s3://*your_user_name* --force

What is the output?

```
website $ aws s3 rb s3://carrd --force
delete: s3://carrd/error.html
delete: s3://carrd/hello.html
remove_bucket: carrd
website $
```

Submission instructions

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.